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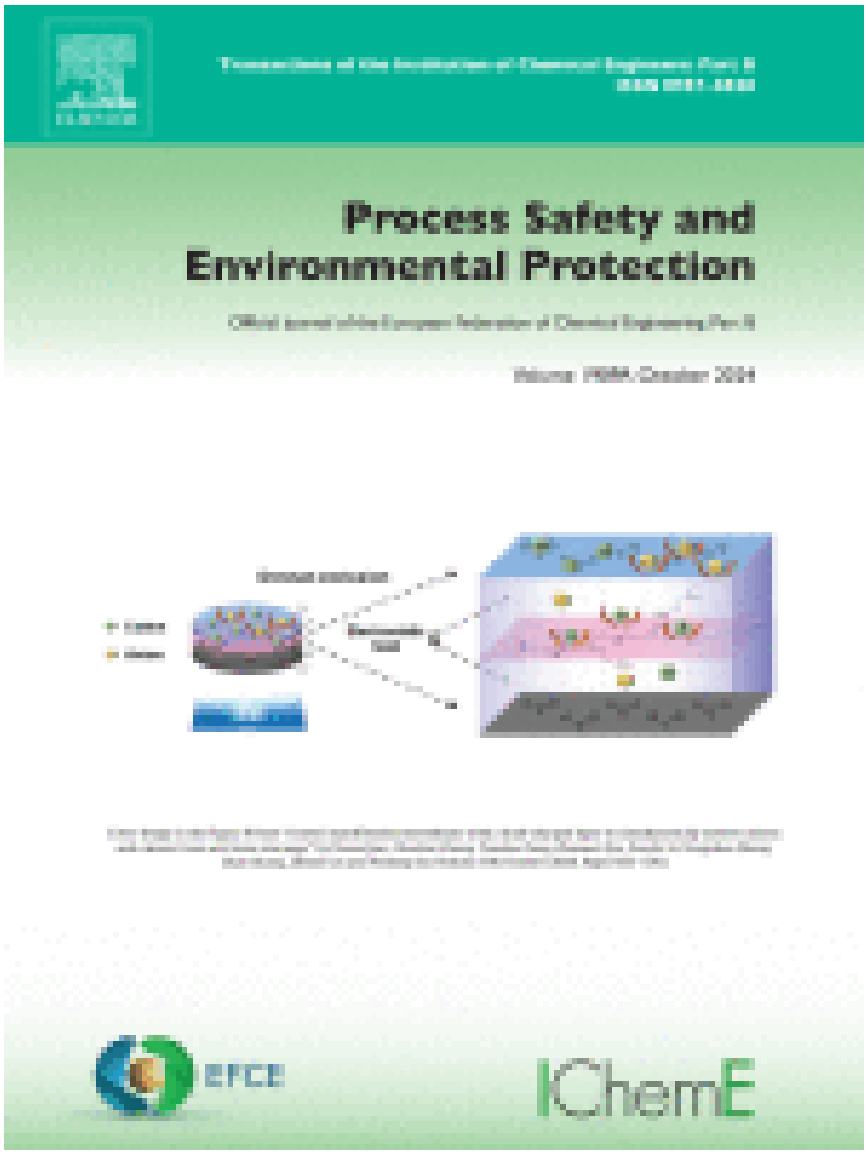
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Influence of corrugated tube structure and flow rate on the hydrogen absorption performance of metal hydride reactor and structural optimization

Wenyan Bi [a], Yikai Hou [a], Jianfeng Wan [b,*], Jian Hu [b], Menglin Yu [b], Xiangxuan Xu [b], Yihong Sun [b], Qiang Zhou [a], Shizheng Xie [b], Tianen Li [b], Baozhong Liu [a,*]

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ABSTRACT

The metal hydride bed (MHB) reactor is an effective instrument for storing hydrogen. However, thermal stress causes the cooling tube expansion inside the MHB. The long-term effects will lead to the failure of the cooling tube. To absorb the thermal stress and reduce the deformation of the cooling tubes, this article establishes an MHB reactor with corrugated cooling tubes to study its hydrogen absorption performance. The results prove that some parameters are adversarial and flow rate patterns are complex (such as extreme flow rate and insufficient flow rate) in the hydrogen absorption process of the corrugated tube MHB reactor, which makes it impossible to optimize the structure of the corrugated tube through the gradient method. Under the condition of a fixed corrugated tube volume of 3×10^{-6} m^3 and a cooling water volumetric flow rate of 0.5×10^{-4} m^3/s, the structural parameters of the corrugated tube are optimized by the Monte Carlo algorithm. According to the optimized results (N=8, r_o=4.25 mm, r_i=3.98 mm, d_o=3.17 mm, d_i=4.33 mm), the hydrogen absorption efficiency of the corrugated tube MHB reactor (reaction fraction 0.758) is 1.36 times that of the straight tube (reaction fraction 0.557) at 560 s. Thus, the corrugated tube can reduce thermal stress and improve the heat transfer effect. This research is beneficial in improving long-term stability, enhancing operational efficiency, and reducing the volume and weight of large MHB reactors.

1. Introduction

As a clean energy source, hydrogen is an excellent alternative energy source (Olabi and Mohammad Ali, 2022). The storage of hydrogen plays a vital role in the utilization of hydrogen energy (Mehr et al., 2024; Kumar et al., 2024). Currently, commercial hydrogen storage methods include high-pressure gaseous and low-temperature liquid hydrogen storage, which require high energy consumption and bring safety hazards (Zheng et al., 2021, 2012). However, hydrogen storage alloy has remarkable advantages over them in these aspects (Luo et al., 2024; Şenol et al., 2024).

The impact of hydrogen storage alloys on hydrogen storage performance is crucial, and many scholars are researching this area. Satya et al (Satya Sekhar et al., 2013). compared the hydrogen storage performance of two metal alloys (Mg+30 % $MnNi_5$ and $MnNi_{4.08}Co_{0.2}Mn_{0.62}Al_{0.1}$) in a specific temperature range. To reduce the hydrogen generation barrier of Mn, Ma et al (Ma et al., 2020). and Zhang et al (Zhang et al., 2019). prepared alloys by incorporating TiO_2 through ball milling. $NaAlH_4$ is currently a widely studied coordination hydride, which undergoes three steps of hydrogen release. The high kinetic energy in the first two steps results in a high dehydrogenation temperature, poor hydrogen absorption and limited reversibility of release. Rangsunvigit et al (Rangsunvigit et al., 2013). theoretically analyzed that Ti-based oxides can enhance the cycling performance of $NaAlH_4$. Bogdanovic et al (Bogdanović and Schwickardi, 1997). and Sandlock et al (Sandrock et al., 2002). used β-$TiCl_3$ to improve the hydrogen absorption and desorption performance of $NaAlH_4$, yielding favourable results.

Since the period and investment for developing a novel material are relatively large (Drawer et al., 2024), Liu et al (Liu et al., 2024). held that structural design has the lowest cost and the highest effect. They compared three fin types and concluded that the Inter-Fin and Outer-Fin have more advantages than traditional Inner-Fin. In recent years, some innovative research has been done on hydrogen storage in metal

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Nomenclature

r	radius of corrugated tube, m
r_i'	minimal flow radius of corrugated tube, m
r_o'	maximal flow radius of corrugated tube, m
d	diameter of corrugated tube, m
N	number of corrugations
p	pressure, Pa
p_{eq}	equilibrium pressure of hydrogen, Pa
p_{ref}	reference pressure, Pa
T	temperature, K
T_{ex}	temperature of external environment, K
t	reaction time, s
$\boldsymbol{u}$	velocity vector, m/s
m	hydrogen mass reaction rate, kg/(m^3·s)
C_p	specific heat, J/(kg·K)
X	average reacted fraction of metal hydride bed
x	reacted fraction at some point
V_s	volume of metal hydride, m^3
V_c	volume of single corrugated tube, m^3
K	permeability, m^2
R	general gas constant, J/(mol·K)
R_g	hydrogen gas constant, J/(kg·K)
k	thermal conductivity, W/(m·K)
C_a	reaction rate constant, 1/s
C_d	reaction rate constant, 1/s
E	activation energy, J/mol
$\boldsymbol{n}$	normal vector
h_{air}	natural convection heat transfer coefficient, W/(m^2·K)
$\boldsymbol{q}$	symmetric boundary heat flux of MHB, W/m^2
$\boldsymbol{q_0}$	heat flux between MHB and environment, W/m^2
k_m	hydrogen absorption ratio coefficient
Q_f	volumetric flow rate of heat transfer fluid, m^3/s
k_e	turbulent kinetic energy
ν_f	kinematic viscosity of the fluid, m^2/s

Greek Symbols

ρ	density, kg/m^3
θ	porosity
μ	dynamic viscosity, kg/(m·s)
ΔH	enthalpy of formation, J/mol
ε	turbulence energy dissipation rate, m^2/s^3
σ_k	turbulent kinetic energy Prandtl number
σ_ϵ	turbulent dissipation rate Prandtl number

Subscripts

g	hydrogen
s	metal hydride bed
f	heat transfer fluid
eff	effective value
in	inlet
out	outlet
i	inner arc
o	outer arc
sat	saturation
emp	empty
a	absorption
d	desorption
w	metal hydride reactor wall
0	initial value

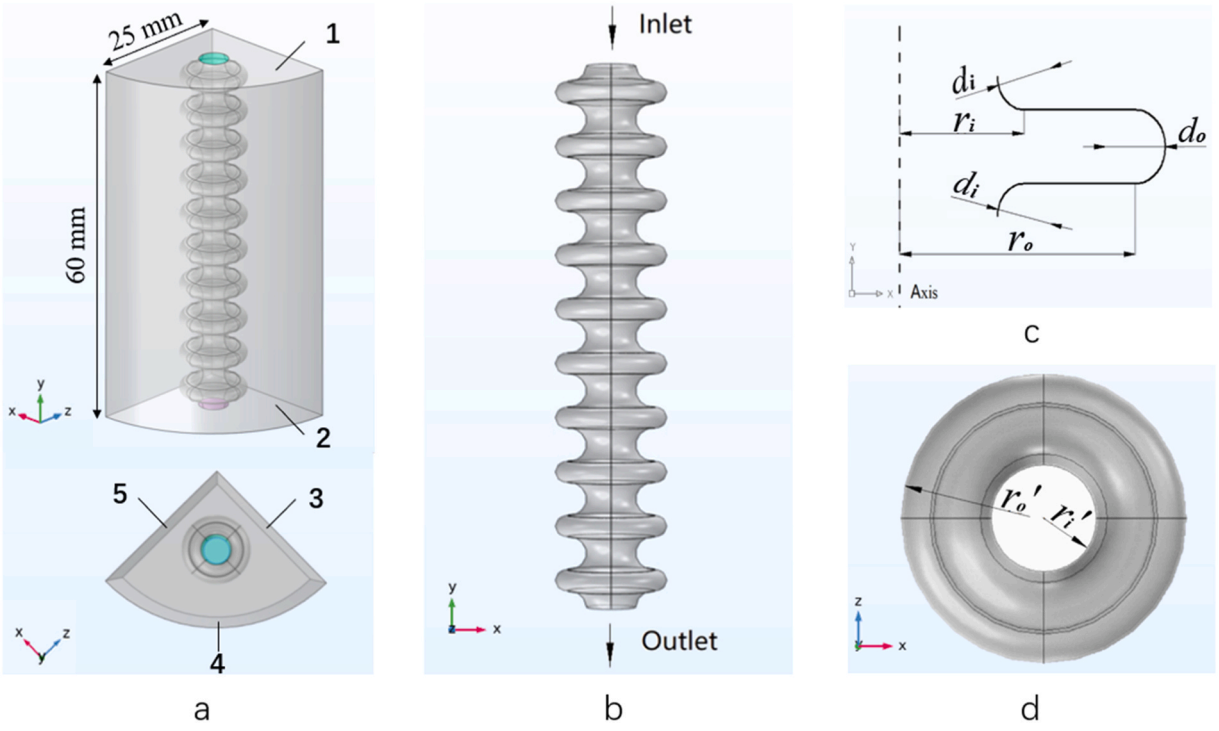


Fig. 1. Physical model of the MHB and corrugated tube. (a) The quarter physical model and its surface definition; (b) Three-dimensional graphics of the corrugated tube; (c) The two-dimensional slice of a single corrugated tube (The inner and outer arc diameters are d_i and d_o, respectively). The distance between the inner arc's (the outer arc's) center and the axis is r_i (r_o); (d) The top view of a single three-dimensional corrugation (r_i' is minimal flow radius, and r_o' is maximal flow radius).

hydride. For example, Muhammad Aziz's team (Lesmana and Aziz, 2023; Lesmana et al., 2023) employed metal hydride with triply periodic minimal surface structures to integrate metal beds with the frame of cars, thus improving the structural strength and heat transfer efficiency. They conducted experimental research and achieved remarkable results.

When applying metal hydrides to fuel vehicles, it is necessary to reduce the transient and long-term damage caused by thermal stress. Davids et al (Lototskyy et al., 2020). added an appropriate amount of expanded natural graphite to decrease the thermal stress generated by metal hydride. Ma et al (Ma et al., 2021). considered that hydrogen storage alloy particles expand due to the release of heat (Joubert et al., 2002), become brittle under the action of hydrogen, and are easy to crack and turn into powders (Yehui et al., 2022; Jing et al., 2021). Affected by gravity, these powders undergo sinking and compaction. The long-term operation will generate massive stress at the bottom (Lee et al., 2022), especially when the hydrogen storage tank is large. However, hydrogen transportation and photovoltaic or wind energy production require larger and cheaper hydrogen storage tanks (Drawer et al., 2024; Ren et al., 2023). Therefore, as the size of the hydrogen storage tank increases, cooling tubes must have better heat exchange performance and can resist local deformation caused by gravity and thermal stress. Various deformations may occur even when adopting coiled tubes due to temperature rise (Fan et al., 2022).

Based on the rapid development of hydrogen storage tanks and their potential hazards, it is necessary to develop a cooling tube capable of absorbing thermal stress and exhibiting excellent heat exchange performance. The corrugated tube structure has unique characteristics, which can effectively absorb axial and transverse deformation (Fuqiang et al., 2016; Yang et al., 2020). Regarding heat transfer, Selimefendigil et al (Selimefendigil and Öztop, 2020, 2019). conducted a study on magnetohydrodynamics forced convection of nanofluid. They found that a U-shaped ventilated cavity can effectively enhance the wall heat transfer coefficient. After further investigation, they summarized the influence of corrugated partition parameters on convection. Some studies (Selimefendigil and Öztop, 2020; Nguyen et al., 2024; Yu et al., 2024) have also indicated that corrugated tube structure helps improve heat transfer performance. This article investigates the effects of structural parameters and flow rate on hydrogen absorption performance, which is beneficial to improving the operational stability and hydrogen absorption efficiency of large hydrogen storage tanks.

2. Numerical model

The model is an MHB reactor embedded with a corrugated tube, where the filling material is $LaNi_5$. Heat transfer fluid (water) in the tube removes the heat from the MHB to promote hydrogen absorption. To simplify the calculation, the authors develop the following assumptions:

(i) Radiative heat transfer is not considered among all the heat transfer processes.

(ii) Hydrogen is non-viscous when flowing in the MHB without considering the effect of dissipation on temperature.

(iii) The properties of metal hydride do not vary with temperature and concentration (Bao et al., 2013).

(iv) The heat transfer fluid (water) is assumed incompressible.

(v) The local heat balance is effective in the MHB reactor.

2.1. Model establishment

Fig. 1 depicts the physical model of the corrugated tube and the MHB reactor. To conveniently compare the corrugated tube structural effects on heat transfer performance, set all the corrugated tubes at the same volume (3×10^{-6} m^3) and $d_i + d_0 = \frac{0.06}{N}$.The equation of corrugated tube volume (V_c) is as follows:

$$V_c = \left[\pi {r_o}^2 d_o + \pi {r_i}^2 d_i + \frac{1}{4}(\pi d_o)^2\left(r_o + 2\frac{d_o}{3\pi}\right) - \frac{1}{4}(\pi d_i)^2\left(r_i - 2\frac{d_i}{3\pi}\right)\right]N \quad (1)$$

The MHB reactor includes the corrugated tube with flowing fluid for heat exchange and the MHB filled with porous metal alloy $LaNi_5$ and hydrogen. Two parts follow respective governing equations (continuity, momentum, and energy).

2.2. Governing equations for MHB

The continuity equation for hydrogen in the MHB:

$$\theta\frac{\partial \rho_g}{\partial t} + \nabla \bullet (\rho_g \boldsymbol{u_g}) = -m \quad (2)$$

where θ is the porosity of the MHB, ρ_g is the density of hydrogen, t is time, $\boldsymbol{u_g}$ is the velocity vector of hydrogen, and m is the mass hydrogen reaction rate (hydrogen mass absorbed unit volume and time).

The continuity equation for metal hydride in the MHB:

$$(1-\theta)\frac{\partial \rho_s}{\partial t} = m \quad (3)$$

where ρ_s is the density of metal hydride.

The ideal gas state equation is valid for hydrogen (Sakoda et al., 2010), and its density is defined as:

$$\rho_g = \frac{p_g}{R_g T} \quad (4)$$

where p_g is the hydrogen pressure and R_g is hydrogen gas constant.

The hydrogen gas velocity can be calculated by Darcy's law (Chung and Ci-Jyun, 2009) because the MHB is porous:

$$\boldsymbol{u_g} = -\frac{K}{\mu_g}\nabla p_g \quad (5)$$

where K is the permeability of the MHB and μ_g is the dynamic viscosity of hydrogen. According to the assumption (ii), hydrogen is non-viscous, namely, μ_g=0. To ensure the meaning of Eq. (5), ∇p_g is equal to zero. That is, the pressure is the same everywhere in the MHB.

The energy equation for the MHB:

$$(\rho C_p)_{eff}\frac{\partial T_s}{\partial t} + \rho_g C_{p,g}\boldsymbol{u_g}\nabla T_s - \nabla \bullet (k_{eff}\nabla T_s) = m(T_s(C_{p,g} - C_{p,s}) - \Delta H) \quad (6)$$

$$(\rho C_p)_{eff} = \theta\rho_g C_{p,g} + (1-\theta)\rho_s C_{p,s} \quad (7)$$

$$k_{eff} = \theta k_g + (1-\theta)k_s \quad (8)$$

where $(\rho C_P)_{eff}$ is the effective volumetric heat capacity of the MHB, k_{eff} is the effective thermal conductivity in the MHB, T_s is the temperature of the MHB, $C_{p,g}$ and $C_{p,s}$ are the specific heat of hydrogen and metal, k_g and k_s are the thermal conductivity of hydrogen and metal, ΔH is the formation enthalpy, and θ is the porosity of the MHB.

Reaction Kinetics: the value of m in Eqs. (2), (3), and (6) can be obtained by Eq. (9).

$$m = \begin{cases} C_a \exp\left(-\frac{E_a}{RT_s}\right)\ln\left(\frac{p_g}{p_{eq,a}}\right)(\rho_{s,sat} - \rho_s), & (p_g > p_{eq,a}) \\ 0, & (p_{eq,a} > p_g > p_{eq,d}) \\ C_d \exp\left(-\frac{E_d}{RT_s}\right)\left(\frac{p_g - p_{eq,d}}{p_{eq,d}}\right)(\rho_s - \rho_{s,emp}), & (p_g < p_{eq,d}) \end{cases} \quad (9)$$

where C_a and C_d are the reaction rate constants of hydrogen absorption and desorption. E_a and E_d are the activation energy of hydrogen absorption and desorption. $\rho_{s,emp}$ and $\rho_{s,sat}$ are the MHB density at empty and saturation. $p_{eq,a}$ and $p_{eq,d}$ are the equilibrium pressure of hydrogen at absorption and desorption.

The equilibrium pressure of the MHB is given by the following

(Chung and Ci-Jyun, 2009):

$$\ln\left(\frac{p_{eq,a}}{p_{ref}}\right) = A - \frac{B}{T_s} \tag{10}$$

where p_{ref} is the reference pressure. A and B are the coefficients in the equilibrium pressure equation for absorption, which are 24.156 and 3704.6, respectively. Due to the hysteresis phenomenon, $p_{eq,a}$ is greater than $p_{eq,d}$. The specific relationship can be described by Eq. (11) (Chung and Ci-Jyun, 2009).

$$\ln\left(\frac{p_{eq,a}}{p_{eq,d}}\right) = \quad 0.13 \tag{11}$$

2.3. Governing equations for the corrugated tube and the heat transfer fluid

This paper chooses the flow heat transfer method of similar structures verified by predecessors to improve the calculation accuracy. Córcolesa et al (Córcoles et al., 2019). conducted a structured optimization simulation studying the flow heat transfer phenomena of corrugated tubes using the Realizable $k-\varepsilon$ model. Before conducting optimization, the experimental model and the simulation result were verified with the literature (Córcoles et al., 2019) (Vicente et al., 2004).

$$\rho_f(\boldsymbol{u_f}\bullet\nabla)\boldsymbol{u_f} = \nabla\bullet\left[-p\mathbf{I} + (\mu_t+\mu_f)(\nabla\boldsymbol{u_f} + (\nabla\boldsymbol{u_f})^{\boldsymbol{T}})\right] \tag{12}$$

$$\rho_f\nabla\bullet\boldsymbol{u_f} = 0 \tag{13}$$

$$\mu_t = \rho_f C_\mu \frac{k_e{}^2}{\varepsilon} \tag{14}$$

where ρ_f is the density of heat transfer fluid, $\boldsymbol{u_f}$ is the velocity of heat transfer fluid, p is the pressure of heat transfer fluid, $\mathbf{I}$ is the tensor, μ_f is fluid viscosity, μ_t is turbulent viscosity, k_e is the turbulent kinetic energy, and ε is the turbulence energy dissipation rate.

The turbulent kinetic energy equation:

$$\rho_f(\boldsymbol{u_f}\bullet\nabla)k_e = \nabla\bullet\left[\left(\mu_f+\frac{\mu_t}{\sigma_k}\right)\nabla k_e\right] + \mu_t[\nabla\boldsymbol{u_f} : (\nabla\boldsymbol{u_f} + (\nabla\boldsymbol{u_f})^{\boldsymbol{T}})] - \rho_f\varepsilon \tag{15}$$

To eliminate the turbulent energy dissipation rate ε in the above equation, add Eq. (16):

$$\rho_f(\boldsymbol{u_f}\bullet\nabla)\varepsilon = \nabla\bullet\left[\left(\mu_f+\frac{\mu_t}{\sigma_\varepsilon}\right)\nabla\varepsilon\right] + C_{\varepsilon1}\frac{\varepsilon}{k_e}\mu_t[\nabla\boldsymbol{u_f} : (\nabla\boldsymbol{u_f} + (\nabla\boldsymbol{u_f})^{\boldsymbol{T}})] - C_{\varepsilon2}\rho_f\frac{\varepsilon^2}{k_e+\sqrt{\nu_f\varepsilon}} \tag{16}$$

where σ_k and σ_ε are the turbulent Prandtl number of kinetic energy and dissipation rate, ν_f represents the kinematic viscosity of the fluid. C_μ, $C_{\varepsilon1}$, and $C_{\varepsilon2}$ are turbulent constants. The coefficient's default value (Versteeg and Malalasekera, 2007) : $C_{\varepsilon1} = 1.44$, $C_{\varepsilon2} = 1.92$, $C_\mu = 0.09$, $\sigma_k = 1.00$, and $\sigma_\varepsilon = 1.30$. The other critical parameter values required in the simulation are listed in Table S1.

Without considering the heat from the flow dissipation, the energy equation of the transfer fluid in the corrugated tube is written as follows:

$$\rho_f C_{p,f}\frac{\partial T_f}{\partial t} + \rho_f C_{p,f}\boldsymbol{u_f}\nabla T_f - \nabla\bullet(k_f\nabla T_f) = 0 \tag{17}$$

where $C_{p,f}$ is the specific heat capacity of the heat transfer fluid, T_f is the temperature of the heat transfer fluid, and k_f is the thermal conductivity of the heat transfer fluid.

2.4. Initial and boundary conditions

For top surface 1:

The inlet pressure of hydrogen on top surface 1 increases with the following curve during the hydrogen absorption:

$$p_{in} = \begin{cases} 0.143 + (0.8-0.143)t/100\ , & (0 < t < 100\text{s}) \\ 0.8\ , & (t \geq \quad 100\text{s}) \end{cases} \tag{18}$$

where p_{in} is the inlet pressure. The hydrogen pressure in the MHB is evenly distributed and equal to the pressure at the entrance.

For bottom surface 2 and side surface 4:

Set up the natural convection heat exchange between the MHB wall and the external environment as:

$$\boldsymbol{q_0} = h_{air}(T_{ex} - T_w) \tag{19}$$

where $\boldsymbol{q_0}$ is the heat flux exchanged by natural convection between the MHB wall and the external environment, T_{ex} is the external environment temperature, T_w is the temperature of the MHB wall, and h_{air} is the coefficient of natural convective heat transfer.

For side surfaces 3, 5:

Set symmetry boundary conditions:

$$-\boldsymbol{n}\bullet\boldsymbol{q} = 0 \tag{20}$$

$$-\boldsymbol{n}\bullet\boldsymbol{u_g} = 0 \tag{21}$$

where $\boldsymbol{n}$ is the normal unit vector of the wall, $\boldsymbol{q}$ is the heat flow intensity, $\boldsymbol{n}\bullet\boldsymbol{q}$ denotes the heat flow intensity through the wall, and $\boldsymbol{n}\bullet\boldsymbol{u_g}$ indicates the velocity through the wall.

The initial metal density ($\rho_{s,0}$) in the MHB reactor is given by:

$$\rho_{s,0} = \quad \rho_{s,emp} \tag{22}$$

The initial metal temperature ($T_{s,0}$) in the MHB reactor is given by:

$$T_{s,0} = T_{ex} \tag{23}$$

The above are the initial and boundary condition settings for the MHB. The condition settings of the fluid inside the corrugated tube are as follows:

To compare the heat transfer effects on different corrugated tubes uniformly, all of the volumetric flow rates at the inlet remain consistent, except for the specifically stated cases, which are set as:

$$Q_f = 2\times10^{-7}\text{m}^3/\text{s} \tag{24}$$

The heat transfer fluid's inlet temperature ($T_{f,in}$) is set to the ambient temperature:

$$T_{f,in} = T_{ex} \tag{25}$$

The heat transfer fluid's outlet temperature ($T_{f,out}$) is set by the adiabatic boundary condition:

$$\frac{\partial T_{f,out}}{\partial \boldsymbol{n}} = 0 \tag{26}$$

The inlet velocity ($u_{f,in}$) of the heat transfer fluid is set as:

$$u_{f,in} = \frac{Q_f}{\pi\left(r_i - \frac{d_i}{2}\right)^2} \tag{27}$$

The outlet pressure ($p_{f,out}$) of the heat transfer fluid is set as:

$$p_{f,out} = 0\text{Pa} \tag{28}$$

Non-slip boundary conditions are adopted for the corrugated tube wall.

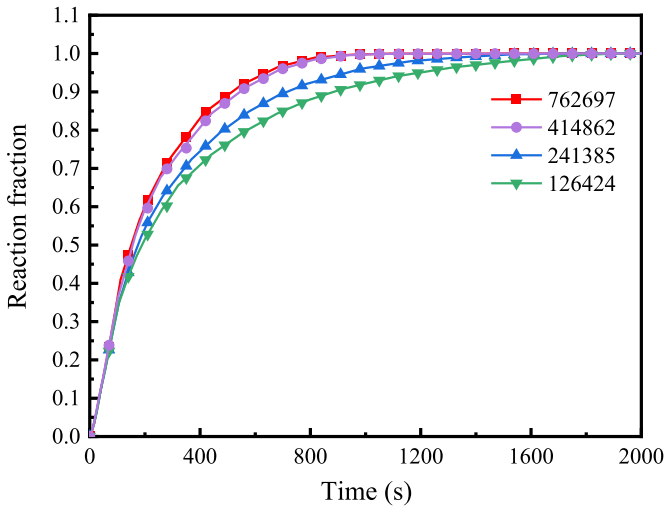


Fig. 2. Grid independence verification diagram of hydrogen absorption process.

2.5. Grid independence validation and model validation

Grid independence validation is performed to obtain accurate numerical results at lower computational costs. All areas in the model use tetrahedral grids, and the internal corrugated tube needs to be encrypted due to its irregularity. Under unchanged boundary layer parameter settings, this article verifies the following four types of grids: 126424, 241385, 414862 and 762697. Set the time step to 0.5 s and the relative tolerance to 0.001. The flow is calculated using a steady-state solver, while the heat transfer is calculated using a transient solver. The momentum and energy equations in the flow calculation adopt a second-order upwind. Firstly, steady-state calculations of the flow are performed, and then the steady-state results are used as initial conditions for transient heat transfer. Calculate the average reaction fraction of the MHB over time for four different grid numbers, as shown in Fig. 2. According to the calculation results of 2000 s, the average reaction fraction for grids with numbers 126424 and 241385 has a significant deviation over time, exceeding 12 %. Compared to the curves obtained for the other two grids, the deviation of the calculation results for grids with numbers 414862 and 762697 is within 5 %. The two curves almost overlap, indicating that the grid does not significantly affect the calculation results. Therefore, considering the computational cost and accuracy requirements, this article uses a grid of 414862 to simulate the

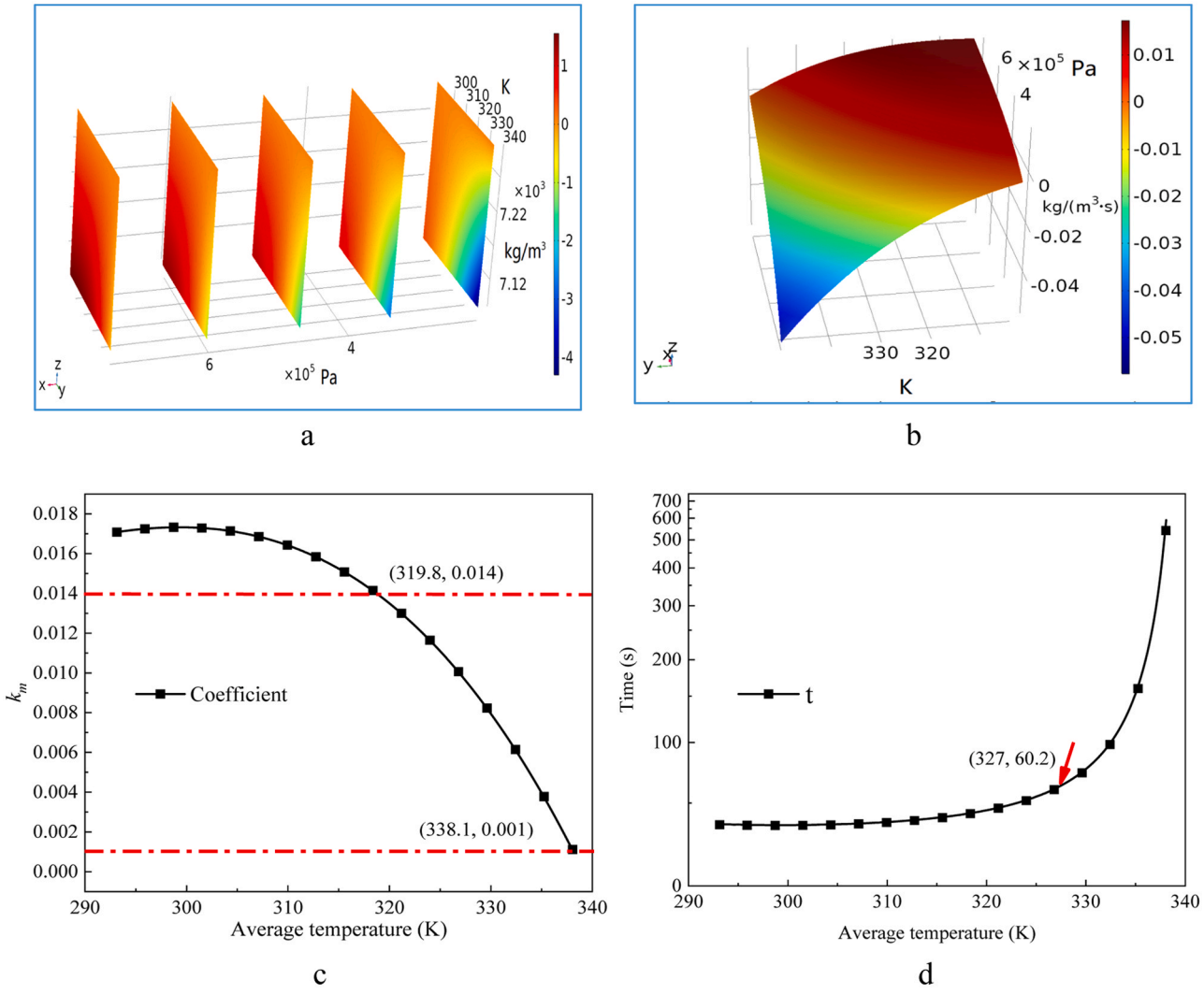


Fig. 3. (a) The relation of hydrogen absorption rate m with p_g, T_s and ρ_s (The x-axis is p_g, the y-axis is T_s, and the z-axis is ρ_s. The color represented the value of m.); (b) The relationship between k_m, p_g, and T_s (The x-axis is p_g, the y-axis is T_s, and the z-axis is k_m.); (c) The relationship between k_m and T_s; (d) The time required for the reaction fraction between 0 and 0.7 at various temperatures when p_g= 0.8 MPa.

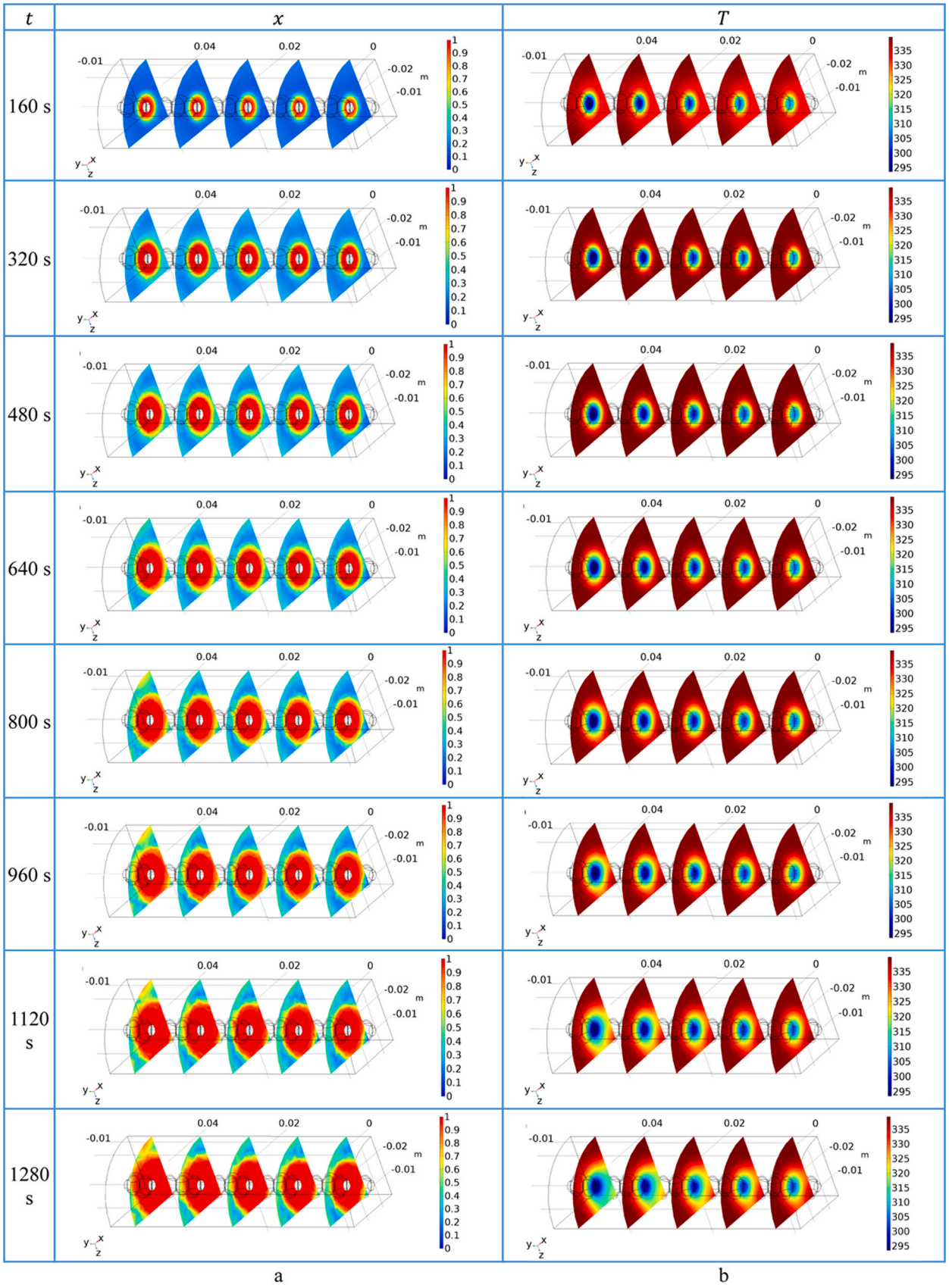


Fig. 4. The diagrams of (a) reaction fraction and (b) temperature distribution at different moments. (N=10, r_0=3.94 mm, r_i=3.5 mm, d_0=3.5 mm, and d_i=3.5 mm).

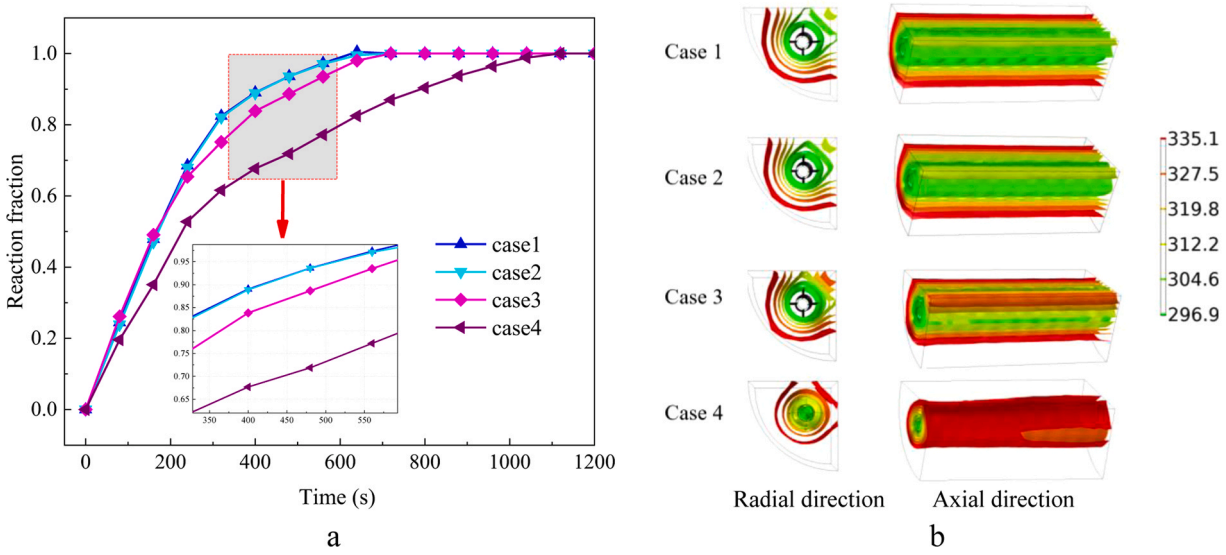


Fig. 5. (a) Simulation results of the average reaction fraction and time at various volumetric flow rates. (b) The isothermal surface diagrams at different volumetric flow rates when $t = 480$ s.

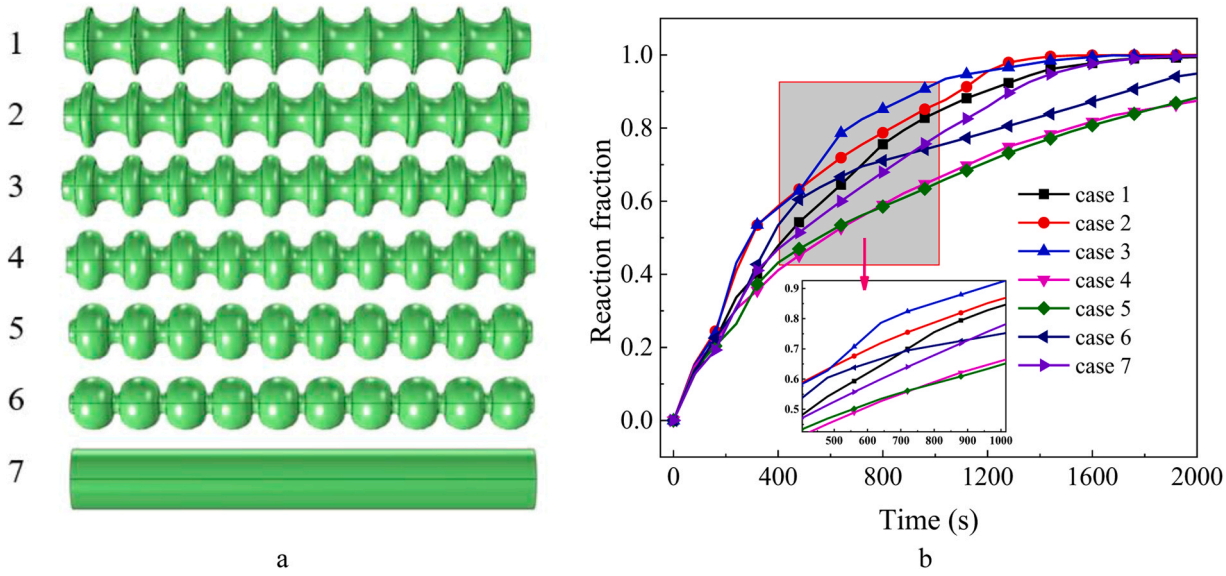


Fig. 6. (a) Geometry models of various corrugated tubes; (b) Simulation result of average reaction fraction and time.

hydrogen absorption of the MHB.

To validate the correctness of the designed computational model, compare the computation results with the experimental results in the literature (Abdelmajid et al., 1999) at the same initial and boundary conditions when Z = 25 and Z = 35 mm (Figure S1). It is found that the trends of the simulation and experimental results are roughly consistent. There are two main differences. Firstly, the simulated temperature slightly exceeds the experimental temperature between 100 s and 600 s. This is because the hydrogen pressure measured in the experiment is at the hydrogen storage reactor's entrance. However, the physical model simulated in this paper ignores the expansion space at the experimental model's entrance. As a result of not accounting for pressure loss at the experiment's reactor's inlet and internal expansion area, the simulated hydrogen pressure in the model slightly exceeds the experimental pressure. Correspondingly, the simulated temperature is slightly higher. Secondly, in the range from 2400 s to 3200 s, the simulated temperature is somewhat lower than the experimental temperature. This is because the hydrogen storage reactor shell used in the experiment has a specific heat capacity when convective heat transfer with the external environment occurs. However, the heat capacity of the shell is not considered in the simulation. According to the above, there is a slight difference between simulated and experimental results, and the trend of temperature curves is nearly consistent, so the model's validity and correctness are well confirmed.

3. Results and discussion

3.1. Analysis of the hydrogen absorption rate equation

The hydrogen absorption rate equation:

$$m = C_a \exp\left(-\frac{E_a}{RT_s}\right) \ln\left(\frac{p_g}{p_{ref}\exp\left(A - \frac{B}{T_s}\right)}\right)(\rho_{s,sat} - \rho_s) \tag{29}$$

The equation of the hydrogen absorption rate coefficient k_m:

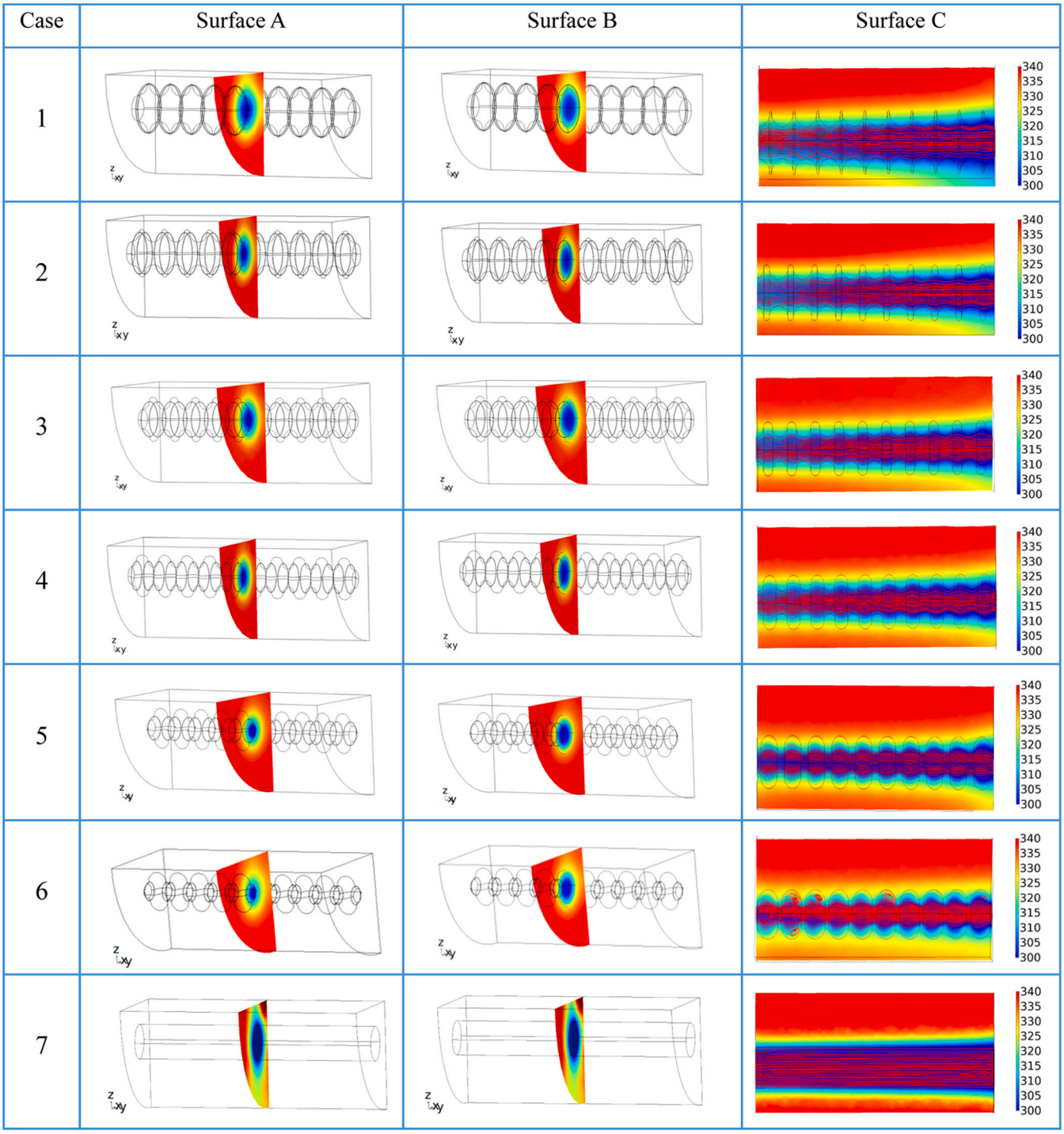


Fig. 7. When $t = 720$ s, the temperature distributions of Surface A, B, and C. (Surface A: the plane passing through the minimal flow in the middle; Surface B: the plane passing through the maximal flow in the middle; Surface C: the plane passing through the center of the corrugated tube and the intersection of two symmetrical planes in the MHB.).

$$k_m = C_a \exp\left(-\frac{E_a}{RT_s}\right) \ln\left(\frac{p_g}{p_{ref}\exp(A - \frac{B}{T_s})}\right) \tag{30}$$

From Fig. 3a, the hydrogen absorption rate can be effectively improved by decreasing T_s, increasing p_g, or decreasing ρ_s. Ignoring the effect of $(\rho_{s,sat} - \rho_s)$ in Eq. (29), m is transformed to k_m. The larger the k_m, the faster the hydrogen absorption. From Fig. 3b, p_g and T_s have similar effects on k_m, but p_g is hard to adjust freely. Modifying T_s is the most effective approach to change k_m and the hydrogen absorption rate m. Lowering T_s can increase k_m, however, this adjusting capacity becomes weaker with T_s decreases until k_m is unchanged.

The relation between k_m and T_s is an upper convex function (Fig. 3c), so its second derivative is less than 0.

$$k_m(T_1) + k_m(T_2) \le 2k_m\left(\frac{T_1 + T_2}{2}\right) \tag{31}$$

Under the same heat flux and $(\rho_{s,sat} - \rho_s)$, the hydrogen absorption

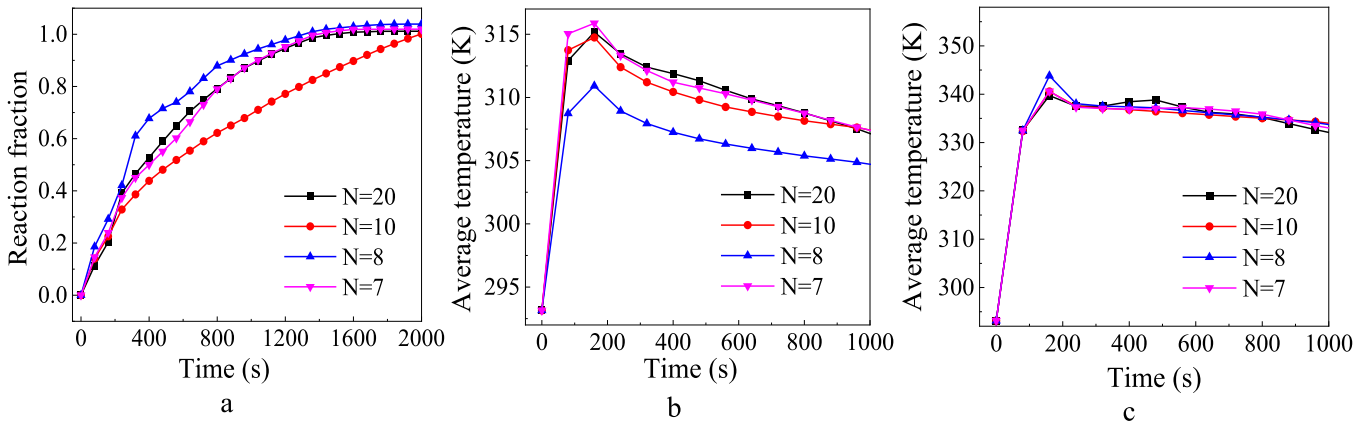


Fig. 8. Simulation results at various *N*. (a) The relation between average reaction fraction and time; (b) The relation between average outlet temperature of heat transfer fluid in the corrugated tube and time; (c) The relation between the average temperature of the MHB and time.

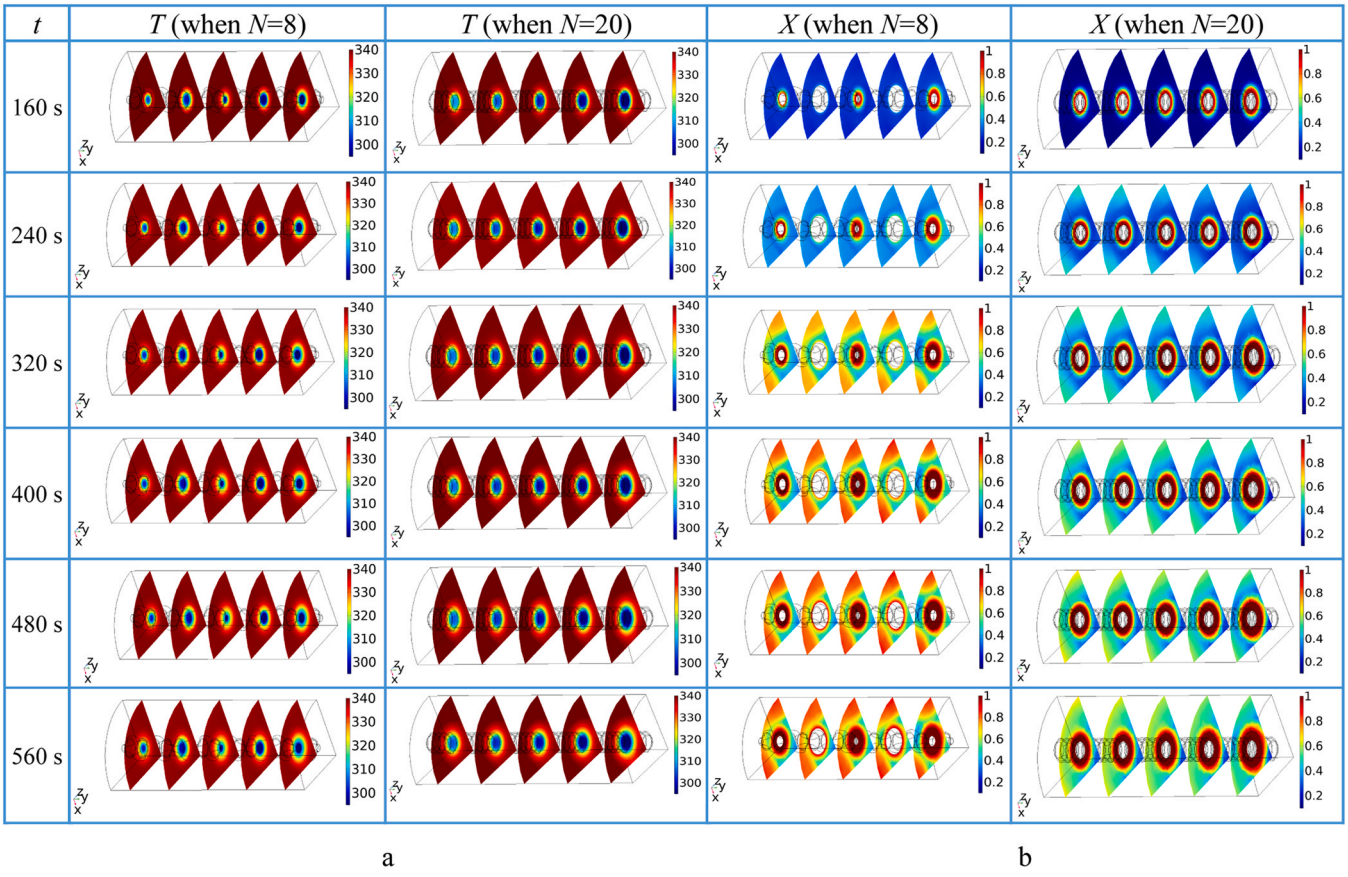


Fig. 9. The comparison of (a) temperature distributions and (b) reaction fraction distributions at different moments.

rate will be the highest when $T_1 = T_2$. Improving uniformity and reducing the highest temperature or the highest temperature area can increase the hydrogen absorption rate. Define x as the reaction fraction at some point in the MHB:

$$x = \frac{\rho_s - \rho_{s,emp}}{\rho_{s,sat} - \rho_{s,emp}} \tag{32}$$

Combine Eqs. (29) and (32), get

$$x = 1 - \exp\left(\frac{1}{\theta - 1} C_a \exp\left(-\frac{E_a}{RT_s}\right) \ln\left(\frac{p_g}{p_{ref}\exp\left(A - \frac{B}{T_s}\right)}\right) t\right) \tag{33}$$

When $T_s < 319.8$ K, $k_m > 0.014$ (Fig. 3d). The value of k_m is enormous at the initial reaction phase, which is more than 82 % of the maximal value (0.017). ρ_s is tiny at the initial reaction phase, which results in a big $(\rho_{s,sat} - \rho_s)$ in Eq. (29). Combining two aspects, the initial hydrogen absorption rate is extremely fast. When $T_s > 338.1$ K, $k_m < 0.001$, hydrogen absorption almost stops. Optimizing the hydrogen absorption rate is vital to avoiding excessive temperature. In addition, the reaction

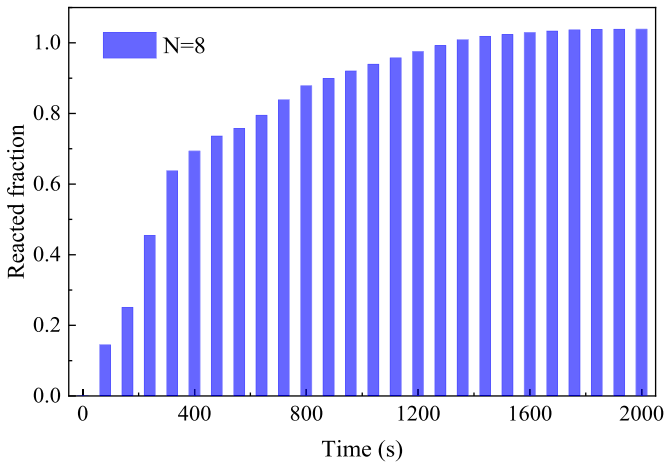


Fig. 10. Optimized simulation results between average reaction fraction and time for the corrugated tube.

time becomes shorter when the temperature of the MHB decreases in Fig. 3d. After that, continuously decreasing the temperature of the MHB will have no better effect on reducing the reaction time.

In Fig. 4a, the blue area indicates that the hydrogen absorption time is prolonged due to the high MHB temperature. The yellow area denotes that hydrogen absorption has not achieved saturation. The red area represents that hydrogen absorption is close to saturation. In Fig. 4b, the color indicates the temperature. The red area in Fig. 4a corresponds to both the yellow and the bright-red regions in Fig. 3b, suggesting temperatures ranging from 305 K to 330 K. The k_m value within this temperature range (305–330 K) varies from 45 % to 100 % of the maximal value (0.017). The temperature in this region is slightly lower, corresponding to an incredibly high hydrogen absorption rate coefficient. This area almost becomes saturated with hydrogen ($x = 1$) immediately. Yellow and blue regions in Fig. 4a overlap with the red-black area in Fig. 4b, the temperature range is from 330 K to 338 K, and the low hydrogen absorption rate coefficient k_m (ranges from 0.0010 to 0.0078) results in a small x value.

From Fig. 4b, the maximal temperature of the MHB is almost the same except at 160 s. And the high-temperature region gradually shrinks with time. This is because when the temperature of the MHB reaches a specific value, the hydrogen absorption rate is very slow, causing minimal heat release. Meanwhile, due to the cooling of the heat transfer fluid, the average temperature decreases continuously until the MHB finally reaches equilibrium. It prevents overheating the MHB persistently.

3.2. Effect of the volumetric flow rate ($\boldsymbol{Q_f}$)

The volumetric flow rate is a decisive factor affecting hydrogen absorption efficiency. It is necessary first to identify an appropriate flow rate condition to compare the performance with various structural parameters.

To better evaluate the hydrogen absorption effect of a whole MHB, the average reaction fraction X is defined as:

$$X = \frac{1}{V_s}\int x dv_s \tag{34}$$

where V_s is the volume of the MHB, and v_s is the micro-volume of the MHB.

Set the geometric parameters of the corrugated tube as follows: $N = 10$, $r_0 = 2.29$ mm, $r_i = 2.25$ mm, $d_0 = 5$ mm, $d_i = 1$ mm. The volumetric flow rate is respectively 1×10^{-4} m^3/s (case 1), 0.5×10^{-4} m^3/s (case 2), 1×10^{-5} m^3/s (case 3), and 0.5×10^{-5} m^3/s (case 4).

From Fig. 5a, the reaction fraction increases with the increase of the volumetric flow rate. However, when the volumetric flow rate exceeds 0.5×10^{-4} m^3/s, the reaction fraction curves nearly coincide. To further analyze the reasons, make the isothermal surface diagrams (Fig. 5b) of the above four flow rates at $t = 480$ s when the X difference is largest. The isothermal surface of case 1, case 2, and case 3 is cylindrical, while case 4 is trumpet-shaped, with a large inlet and a small outlet. In case 4, the temperature difference of the fluid in the corrugated tube is relatively significant. The cooling fluid has enough time to be heated up in the long fluid circulation period, but the heat cannot be quickly carried off due to the insufficient flow rate. Increasing the volumetric flow rate will carry off more heat and rapidly decrease the MHB temperature. In cases 1 and 2, although the volumetric flow rate inflow rate reaches a certain level, the fluid's speed of heat taken away is faster than that of heat generated. Increasing the flow rate does not significantly improve the hydrogen absorption rate in this situation. Thus, a suitable volumetric flow rate of 0.5×10^{-4} m^3/s is selected as the extreme flow rate and adopted in the later investigation.

The above analysis indicates two situations: extreme and insufficient flow rates. When the flow rate approaches the extreme, the isothermal surface takes on a cylindrical shape. When the flow rate is insufficient, the isothermal surface takes on a trumpet shape.

3.3. Effect of the inner and outer arc diameter ($\boldsymbol{d_i}$ and $\boldsymbol{d_o}$)

To investigate the impact of structural parameters (d_i and d_o) on hydrogen absorption rate, according to Table S2, we build six corrugated tubes (case 1 to case 6) and a straight tube (case 7) models with the same volume (Fig. 6a). From Fig. 6b, the heat transfer effect of all cases, including the straight tube, is nearly the same in the first 100 s. But a significant difference arises after 200 s. Notably, the straight tube heat transfer effect here is in the middle.

To further study the heat transfer influencing factors, the time with the largest X difference ($t = 720$ s) in Fig. 6b is selected for comparison. The order of hydrogen absorption performance is as follows: case 3> case 2> case 1= case 6> case 7 >case 4= case 5. The temperature distribution diagrams of three cross sections (Surface A, B and C) at $t = 720$ s are also drawn (Fig. 7). At the entrance of the tube in Surface C, 200 points are uniformly produced, from which 200 flow lines displayed in red line are generated.

Heat transfer area (HTA) is an important factor affecting heat transfer. Compared with the straight tube, the corrugated tube has more surface area. According to Table S2 in Supplementary Materials, the order of HTA is: case 1> case 2> case 3> case 4> case 5> case 6, which is not consistent with the hydrogen absorption performance in Fig. 6b. In particular, the HTA of case 4 and case 5 is larger than that of case 6, but the hydrogen absorption performance is the opposite. By observing the flow lines diagram of Surface C in Fig. 7, it can be seen many secondary vortices in the divergent region of the tube in case 6. Due to the action of secondary vortices, the cooling effect in case 6 is remarkably improved, resulting in a better effect than in case 4 and case 5. Although case 1 has the largest HTA, its temperature distribution in Fig. 7 is the trumpet shape with the largest opening among all cases, which indicates that its flow rate is insufficient. In this scenario, the optimal working condition for case 1 is not achieved, and an increase in flow rate will realize its advantages. The situation in case 2 is similar to that in case 1, but not as obvious. According to Table S2, case 3 has a larger heat transfer area per unit clearance volume (HTAP) than case 1 and case 2. In other words, the axial clearance of case 3 contains less metal alloy. The metal alloy in the clearance preferentially transfers heat radially to the convergent region with a lower temperature, while the divergent region makes it difficult to exchange heat axially. As a result, case 3 has a smaller heat transfer per unit area than case 1 and case 2, leading to a better hydrogen absorption performance. Unlike all other corrugated tubes, the structure of the straight tube has a larger mainstream velocity, which enables it to carry away more heat per unit area (From surface C in

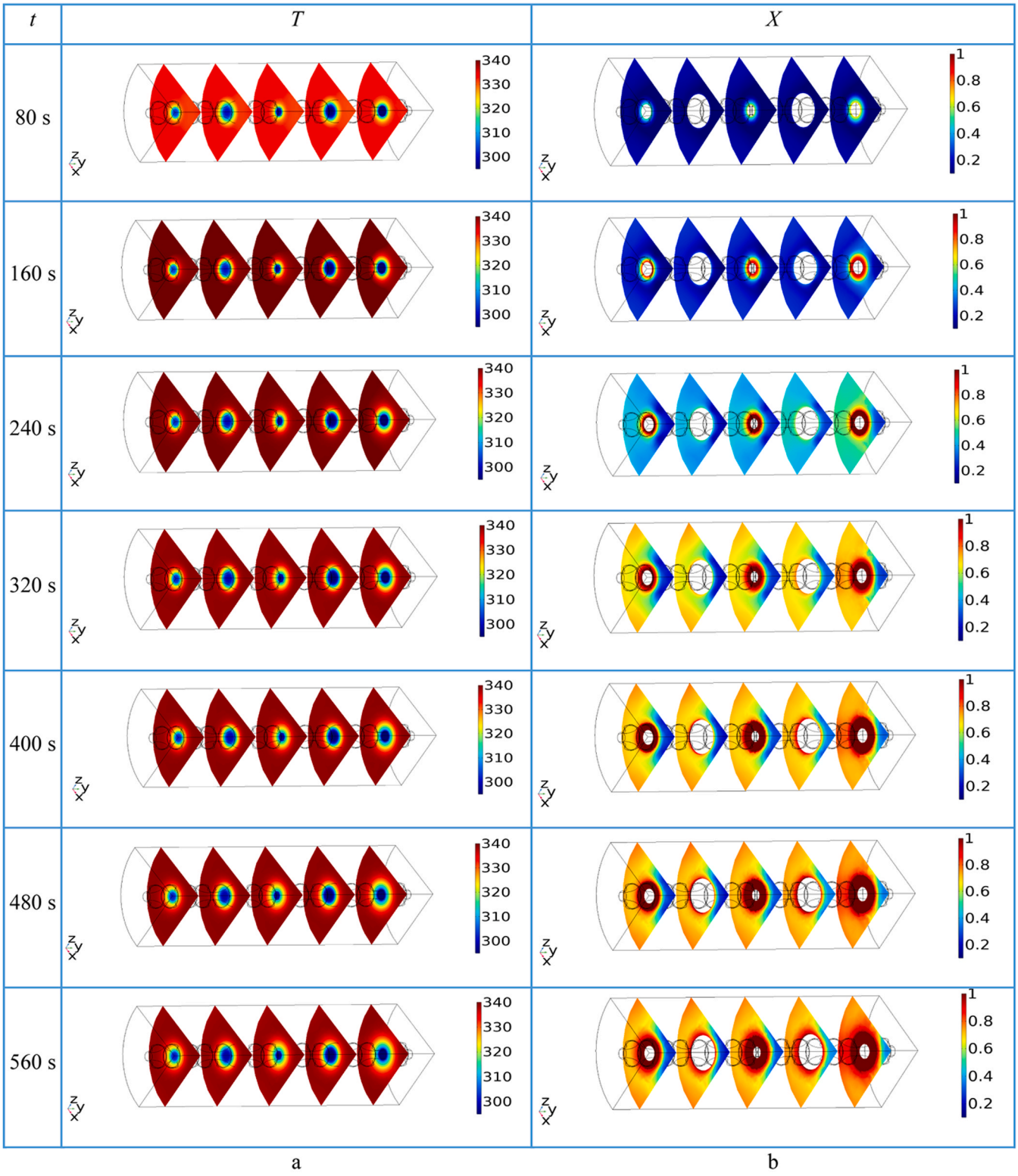


Fig. 11. The diagrams of (a) temperature distribution and (b) reaction fraction distribution at different moments.

Fig. 7, the surface temperature gradient of the straight tube is the largest). Although case 4 and case 5 have a larger surface area than the straight tube, they do not reach the extreme flow rate, resulting in a less effective heat transfer effect.

*3.4. Effect of the corrugations number (**N**)*

The number of corrugations *N* in a corrugated tube is another important parameter influencing heat transfer. According to Table S3, we create four physical models to study the effect of *N* on hydrogen absorption performance.

Fig. 8a shows the hydrogen absorption performance order for

different N: case 3 (N=8)> case 1 (N=20)= case 7 (N=10)> case 4 (N=7). Fig. 8b indicates that, except for N = 8, all average outlet temperatures tend to be the same when the time reaches 1000 s. Combined with Fig. 8c, in the case of N=8, the temperature distribution is more uniform along the radial and axial directions.

The temperature distribution diagrams (Fig. 9) at different moments are compared to verify the above speculation.

From Fig. 9a, the temperature of MHB in the case of N=8 is more uniform at most moments. In other words, when the lowest temperature is the same, the highest temperature is lower, or the high-temperature area is smaller in the case of N=8, which will help raise X.

From Fig. 9b, in most regions, the X value in the case of N=8 is higher than in the case of N=20, particularly in the two rounded corners. Meanwhile, the temperature difference at rounded corners is insignificant (Fig. 9a).

As Table S3 shows, both HTA and HTAP increase as N increases. Due to the coupling of many factors, the case of N =8 has the best performance.

3.5. Optimization

According to the previous analysis, the effect of corrugated tube structural parameters on the hydrogen absorption rate is complex, and the rule can not be obtained through general analysis. At the same time, there may be multiple local extremums. Therefore, the gradient method cannot be utilized to find its extreme value. Here, the Monte Carlo algorithm is adopted for optimization (see Method S1 Monte Carlo algorithm in Supplementary Materials). The final optimization result is as follows: N=8, r_o=4.25 mm, r_i=3.98 mm, d_o=3.17 mm, d_i=4.33 mm, X=0.758.

From Fig. 10, the reaction fraction of the optimized corrugated tube rises rapidly and almost reaches 1.0 in 2000 s

Compared with the straight tube (Fig. 6b), the reaction fraction of the optimized corrugated tube (0.758) is 1.36 times that of the straight tube (0.557) at 560 s, demonstrating that the corrugated tube can significantly improve the hydrogen absorption rate.

Fig. 11 shows the diagrams of temperature distribution and reaction fraction distribution at different moments. From Fig. 11a, the temperature distribution in the optimized MHB is almost unchanged after 400 s, indicating that the heat produced by the MHB reactor and heat removed by the cooling fluid have nearly reached equilibrium. As shown in Fig. 11b, the reaction fraction has reached a high level and is almost unchanged after 400 s.

4. Conclusion

The corrugated tube MHB reactor has good mechanical properties and can absorb volume expansion stress. Compared with the ordinary straight tube, it can improve heat transfer. Through analysis of the volumetric flow rate, structure and optimization, the following conclusions are obtained:

(1) The volumetric flow rate of cooling water has an important impact on the hydrogen absorption performance, and there may be two ultimate situations when the flow rate changes. (i) The extreme flow rate: when the flow rate increases to a specific value, the hydrogen absorption performance will be minimal, and the isothermal surface of the MHB is cylindrical. (ii) The insufficient flow rate: when the flow rate is reduced, the hydrogen absorption performance changes significantly, and the isothermal surface of the MHB is trumpet-shaped.

(2) Reducing the metal bed's temperature difference (axial and radial) helps improve hydrogen absorption. When the flow rate is insufficient, increasing the flow rate effectively reduces the axial temperature difference. Changing the structural parameters of the corrugated tube can reduce the radial temperature difference.

(3) The influence of corrugated tube structural parameters on hydrogen absorption performance is complex, and there are antagonistic parameters: heat transfer area (HTA) and heat transfer area per unit clearance volume (HTAP). Increasing HTA or decreasing HTAP can improve both hydrogen absorption and performance. However, increasing HTA will also increase HTAP. Due to the existence of antagonistic parameters, the arbitrary corrugated tube structural parameters do not have a monotonic effect on the hydrogen absorption performance. Additionally, the internal flow structure of the corrugated tube affects the heat transfer. When the expansion joint (divergent section) is relatively large, secondary vortices will be generated, which can strengthen the heat transfer efficiency.

(4) Because of the complex influence of structural parameters on hydrogen absorption performance, the optimal value cannot be obtained by the gradient method. In this paper, the Monte Carlo algorithm is adopted to solve better structural parameters (N, r_o, r_i, d_o, d_iWith 3 degrees of freedom) with the hydrogen absorption performance at 560 s as the solution objective. The optimized result demonstrates that the hydrogen absorption performance at 560 s is 1.36 times that of the ordinary straight tube.

The above research results are beneficial in enhancing long-term operation reliability for large MHB reactors, reducing their weight and volume, and improving operation safety, economic efficiency, and hydrogen storage efficiency in industries.

CRediT authorship contribution statement

Yikai Hou: Visualization, Validation. **Wenyan Bi:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Jian Hu:** Writing – review & editing, Writing – original draft, Conceptualization. **Jianfeng Wan:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Formal analysis, Conceptualization. **Xiangxuan Xu:** Software. **Menglin Yu:** Visualization, Software. **Yihong Sun:** Software. **Shizheng Xie:** Data curation. **Qiang Zhou:** Resources. **Baozhong Liu:** Project administration, Funding acquisition. **Tianen Li:** Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.psep.2024.07.028.

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